Sliding Critical Band in RoPE-based Length Extrapolation

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Abstract

Extending the context window of RoPE-based Large Language Models is a central problem, as performance degrades once the inference length exceeds the training length. The prevailing explanation attributes this degradation to low-frequency rotary dimensions rotating into out-of-distribution (OOD) angles beyond a fixed *critical dimension*. We show that this static picture is incomplete: through controlled interpolation experiments, we find that a low-frequency dimension entering an OOD angle is neither sufficient nor necessary for model collapse. Some high-frequency dimensions that complete full rotations during training still require interpolation, while some OOD low-frequency dimensions are performance-neutral. Building on this, we introduce the **Sliding Critical Band**: the set of dimensions that must be interpolated is not fixed in the low-frequency region but *migrates across the spectrum* as the extrapolation ratio grows. We further propose **Spectrum Bandwidth Exhaustion**, which explains why raising the RoPE base improves extrapolation—a larger base lowers the frequency floor, delaying the point at which the sliding band runs off the finite dimension spectrum. Experiments on synthetic string-copying tasks and on LLaMA-7B with the C4 dataset confirm that the band slides consistently across model scales and extrapolation ratios.

1 Introduction

Rotary Positional Encoding (Su et al., 2023) is widely adopted in transformer-based (Vaswani et al., 2017) Large Language Models (LLMs) for natural language processing tasks. However, model performance degrades during inference when the sequence length exceeds the maximum length encountered during training. Meanwhile, in real-world scenarios, large language models frequently encounter texts of considerable length. This challenge renders the enhancement of large language models' context extrapolation capabilities an urgent problem that demands resolution.

Prior works proposed several methods to overcome this problem: Chen et al. (2023) and kaiokendev (2023) pioneered the extension of context windows by introducing Position Interpolation (PI), which rescales the relative distance between tokens to fit within the pre-trained range, followed by fine-tuning on limited long-context data. Departing from this approach, more sophisticated schemes like NTK-aware interpolation (bloc97, 2023a) and YaRN (Peng et al., 2023) have been developed.

The prevailing view holds that the success of extending the long-context window in RoPE-based transformers stems from avoiding low-frequency dimensions entering out-of-distribution (OOD) angles during long-context inference (Liu et al., 2024). However, in this paper, we demonstrate that existing explanations (Peng et al., 2023; Liu et al., 2024) for RoPE's poor extrapolation capability cannot fully predict the model's performance under mild extrapolation. Our analysis reveals a mismatch between existing OOD theories and empirical observations: During long-context inference, low-frequency rotary dimensions rotating into OOD angles are neither a sufficient nor a necessary condition for model collapse.

To bridge the gap between empirical observations and current theories, we introduce the ***Sliding Critical Band*** perspective, shifting from static periodic boundaries to a dynamic spectral window. Our contributions can be summarized as follows:

- **Re-evaluating Periodic Boundaries:** We discover that certain dimensions that complete a full rotation during pretraining harbor hidden threat: without interpolation, they can cause model collapse. Conversely, some dimensions that enter into out-of-distribution angles during inference are actually benign and performance-neutral.

- **Sliding Critical Band:** We formalize a dynamic framework where the dimensions requiring interpolation migrate across the spectrum with the proportion of extrapolation ratio.

- **Spectrum Bandwidth Exhaustion:** Our analysis reveals the mechanism underlying RoPE's base scaling: an increased RoPE base postpones the exhaustion of the available spectrum bandwidth, thereby enhancing the model's capacity for long-context extrapolation.

2 Background

2.1 Rotary Positional Encoding (RoPE)

Su et al. (2023) incorporate positional information to attention mechanism by rotating $\mathbf{q} \in \mathbb{R}^d$ and $\mathbf{k} \in \mathbb{R}^d$ vectors where d is an even number. RoPE splits a d-dimensional vector into $d/2$ subvectors. For each subspace, the transformation is as follows:

$$(\mathbf{q}_i^m)' = \begin{bmatrix} \cos(m/\theta_i) & -\sin(m/\theta_i) \\ \sin(m/\theta_i) & \cos(m/\theta_i) \end{bmatrix} \mathbf{q}_i^m \quad (1)$$

where $\mathbf{q}_i^m \in \mathbb{R}^2$ is the m-th query $(0 \le m \le L-1)$ and the i-th subspace $(0 \le i \le \frac{d}{2}-1)$, $\theta_i = \theta^{\frac{-2i}{d}}$. θ is RoPE base, L is sequence length, d is embedding dimension. The same transformation is applied to $\mathbf{k}^m$ in the preceding steps. For simplicity, we denote the RoPE transformation for i-th subvector $\mathbf{x} \in \mathbb{R}^2$ as a function of $\mathfrak{R}_i(\mathbf{x}, m)$. Then for any vector $\mathbf{v} \in \mathbb{R}^d$ at position m,

$$\mathbf{v}' = \begin{bmatrix} \mathfrak{R}_0(\mathbf{v}_{1:2}, m) \\ \mathfrak{R}_1(\mathbf{v}_{3:4}, m) \\ \vdots \\ \mathfrak{R}_{\frac{d}{2}-1}(\mathbf{v}_{d-1:d}, m) \end{bmatrix} \in \mathbb{R}^d \quad (2)$$

where $\mathbf{v}_{j:k}$ denotes the sub-vector of $\mathbf{v}$ from the j-th to the k-th components, for all $j, k \in [1, d]$.

2.2 Related Works

Context Extension in LLMs To extend the context window of pre-trained LLMs, Position Interpolation (PI) (Chen et al., 2023) linearly scales position indices, though it struggles with high-frequency information loss. Subsequent NTK-aware (bloc97, 2023a) methods, including NTK-by-parts (bloc97, 2023b) and Dynamic-NTK (emozilla, 2023), utilize non-uniform frequency scaling to preserve resolution in high-frequency dimensions. YaRN (Peng et al., 2023) refines this by combining non-uniform scaling with attention temperature adjustment to mitigate entropy increases. For non-linear scaling, LongRoPE (Ding et al., 2024) employs evolutionary search to identify optimal non-uniform interpolation factors, while CLEX (Chen et al., 2024) models length scaling as a continuous dynamical system using Ordinary Differential Equations (ODEs). Prior studies (Xiong et al., 2024; Rozière et al., 2024; bloc97, 2023a) found that increasing the base wavelength of RoPE alleviates the attention decay. (Barbero et al., 2025) observed that lower frequencies are less affected by the relative distance and proposed p-RoPE to improve model's long-context capabilities by replacing the lowest frequencies of RoPE with No Positional Encoding (NoPE) (Haviv et al., 2022; Kazemnejad et al., 2023).

Theoretical Analysis of RoPE Theoretical analysis identifies RoPE's efficacy and limits through architectural and periodic lenses. Slash-Dominant Heads (Cheng et al., 2026) emerge from low-rank queries and cone-shaped embeddings, where the slash attention pattern is crucial in long-context inference (Xu et al., 2025; Jiang et al., 2024). From a periodic perspective, the critical dimension (Liu et al., 2024) determines a threshold for the feature dimension, beyond which extrapolation fails due to transitioning into out-of-distribution (OOD) scenarios. Men et al. (2024) concluded that RoPE's base frequency fundamentally limits the context length; falling below this lower bound results in the model exhibiting "superficial" capabilities, where model maintains low perplexity but its actual long-range information retrieval performance deteriorates.

3 Preliminaries and Experimental Setup

In this section, we describe the experimental setup, including the model architecture, data generation, training, interpolation methodology, and evaluation metric.

Models. We adopt the LLaMA architecture with 4 layers, 2 attention heads, and an embedding dimension of 384 with a total of 7.1M parameters. We use Rotary Positional Encoding(Su et al., 2023) with RoPE base $\theta = 10,000$ except specified in Section 5. Character-level tokenization is used, the vocabulary size is 14, which contains 0-9, =, and special tokens [BOS], [EOS], [PAD].

Task. We primarily focus on string-copying task.

Table 1: Examples of string-copying task

Input (Q: prompt, A: label)		Task difficulty
Q: 12345=	A: 12345	Length of prompt
Q: 123456789=	A: 123456789	

The task is to exactly replicate the input sequence. The example is provided in Table 1.

Data Generation. We follow the pipeline proposed in Lee et al. (2025). We generate the training data $\mathcal{D}_{\text{train}}$ of up to a fixed string length l_0 by uniformly sampling the string length $1 \leq l \leq l_0$ and generating a random sequence with length l. Denoting the input as x_i, labels as y_i,

$$\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}, \quad \text{where Length}(x_i) \leq l_0.$$

Training and Evaluation. During pretraining, we use standard next-token-prediction loss. See more training details in Appendix A. During inference, we employ greedy decoding to generate completions. We employ both exact-match accuracy and perplexity (PPL) to assess model performance. While exact-match accuracy provides a direct measure of task success, it often fails to provide an informative signal in challenging regimes where the model has not yet mastered the task, resulting in a "zero-accuracy" plateau. However, a null accuracy does not imply identical model capabilities. To capture the fine-grained progression of model learning, we use perplexity as a continuous proxy. Perplexity remains sensitive to improvements in the model's internal representations even when the discrete accuracy remains at zero.

Interpolation Methods. To investigate the impact of different frequency components on extrapolation performance, we generalize the standard Position Interpolation framework (Chen et al., 2023). Specifically, instead of uniformly scaling all dimensions, we apply interpolation only to a pre-defined subset of frequency subspaces.

Let $\mathcal{S}$ be the set of indices that are targeted for interpolation and $\mathcal{S}$ is a subset of $\{0, 1, \ldots, \frac{d_{\text{max}}}{2} - 1\}$. For a sequence length L' larger than the maximum training sequence length L_{train}, we modify the transformation function $\mathfrak{R}_i$ for each subspace as follows:

$$\mathfrak{R}_i^*(\mathbf{x}, m) = \begin{cases} \mathfrak{R}_i(\mathbf{x}, m/s) & \text{if } i \in \mathcal{S} \\ \mathfrak{R}_i(\mathbf{x}, m) & \text{if } i \notin \mathcal{S} \end{cases}, \quad \text{where } s = \frac{L'}{L_{\text{train}}} \tag{3}$$

where $\mathfrak{R}_i$ is the original rotary transformation, and s is the extrapolation ratio. Note that this selective interpolation is applied consistently to all query vectors $\mathbf{q}$ and key vectors $\mathbf{k}$ across all positions m and the smaller index corresponding to higher frequency. By strategically choosing the index set $\mathcal{S}$ (e.g., targeting specific frequencies), we can isolate and analyze the sensitivity of the model's extrapolation capability to different spectral components of the positions encoding.

4 Sliding Critical Band

In this section, we will re-examine the common claim regarding the failure mechanism of Rotary Positional Embedding (RoPE) extrapolation. Current literature (Peng et al., 2023; Liu et al., 2024) attributes RoPE extrapolation failure to dimensions that fail to complete a full rotation cycle during training. This boundary is formalized by the critical dimension (d_{extra}), a theoretical threshold introduced by Liu et al. (2024) to partition the spectrum based on training-time periodicity:

$$d_{\text{extra}} = \left\lceil \frac{d_{\text{max}}}{2} \log_\theta \left(\frac{L_{\text{train}}}{2\pi} \right) \right\rceil \tag{4}$$

where d_{max} is head dimension, L_{train} is the longest training sequence length, and θ is RoPE-base. [1] This threshold identifies the first rotary plane where the wavelength exceeds the training sequence length. Consequently, subspaces with indices greater than d_{extra} cover only a partial rotation during training, causing them to enter an out-of-distribution (OOD) state during long-context inference.

While the OOD hypothesis explains performance collapse under extreme extrapolation, it proves insufficient for mild extrapolation (e.g., $s = 1.1, 1.2, 1.5$). To systematically identify which frequencies require interpolation, we employ a truncated interpolation strategy. We define the set of

[1] Note that unlike the standard derivation (Liu et al., 2024), we omit the leading factor of 2 to define d_{extra} as the index of the rotary subspace (i.e., the pair of dimensions) rather than the absolute dimension index.

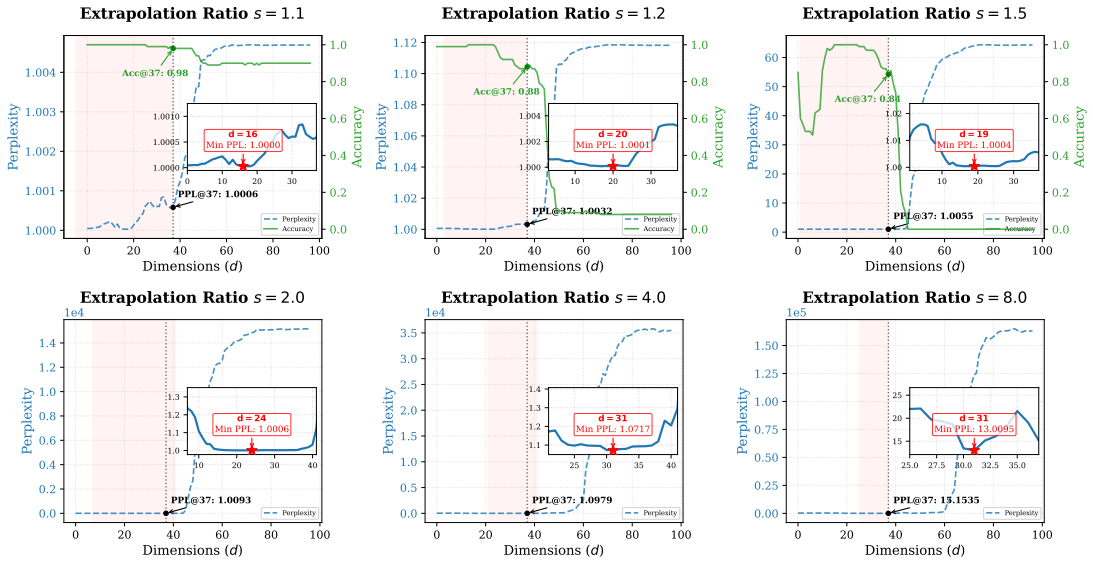


Figure 1: Exclusive-based results of PPL and Accuracy for various extrapolation ratios s for 100-digit string-copying task. Perplexity (PPL) as a function of the interpolation starting index d. The x-axis d represents the upper bound of the dimension range $[d, d_{\text{max}}/2 - 1]$ where interpolation is applied (Equation 5). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

interpolated indices $\mathcal{S}_d$ as:

$$\mathcal{S}_d = \{i \mid d \leq i \leq d_{\text{max}}/2 - 1\}, \quad d \in [0, d_{\text{max}}/2] \tag{5}$$

By increasing the starting index d, we progressively remove high-frequency dimensions from the interpolation set. In our 100-digit experimental setting ($L_{\text{train}} = 203, d_{\text{max}} = 192, \theta = 10000$), the theoretical threshold is $d_{\text{extra}} = 37$.

However, as the first row in Figure 1 shows, under mild extrapolation ratio, the model performance drops before d_{extra}. These results imply that even though the high-frequency dimensions complete the full rotation during training, their OOD state can still have a significant impact on model's extrapolation performance. This finding implies that the OOD state of low-frequency dimensions may not be the complete explanation for extrapolation failure.

In light of these findings, we propose a more dynamic perspective termed ***Sliding Critical Band***, which argues that the dimensions requiring interpolation are not fixed in the low-frequency region, but rather shift across the entire spectrum as the extrapolation ratio s changes.

4.1 Formalizing the Critical Band

To formally define the Sliding Critical Band, we characterize it by two spectral boundaries: the Upper Bound (d_{upper}) and the Lower Bound (d_{lower}). We propose a two-step empirical search procedure: an exclusive-based search for the Upper Bound and an inclusive-based search for the Lower Bound. First, we identify the Upper Bound (d_{upper}) by progressively excluding the high-frequency dimensions from the interpolation set $\mathcal{S}$ as defined in Equation 5. As illustrated in Figure 1, we observe a "U-shaped" performance curve. Interestingly, interpolating the extreme high-frequency dimensions actually results in sub-optimal performance. Under extrapolation ratio $s = 1.2$, the model achieves its peak performance at $d = 20$, which we define as the Upper Bound d_{upper}. Second, fixing the starting index at d_{upper}, we expand the interpolation range by progressively including the lower-frequency dimensions. The set of interpolation indices $\mathcal{S}$ is a function of the threshold d:

$$\mathcal{S}_d = \{i \mid d_{\text{upper}} \leq i \leq d\}, \quad d \in [d_{\text{upper}} - 1, d_{\text{max}}/2 - 1] \tag{6}$$

Figure 2 shows that as more dimensions are added to $\mathcal{S}$, the PPL decreases until it reaches a stable plateau. At $s = 1.2$, this convergence occurs at $d = 68$. We define this saturation point as the Lower Bound (d_{lower}), representing the limit beyond which further interpolation becomes redundant.

Based on these identified boundaries, we can

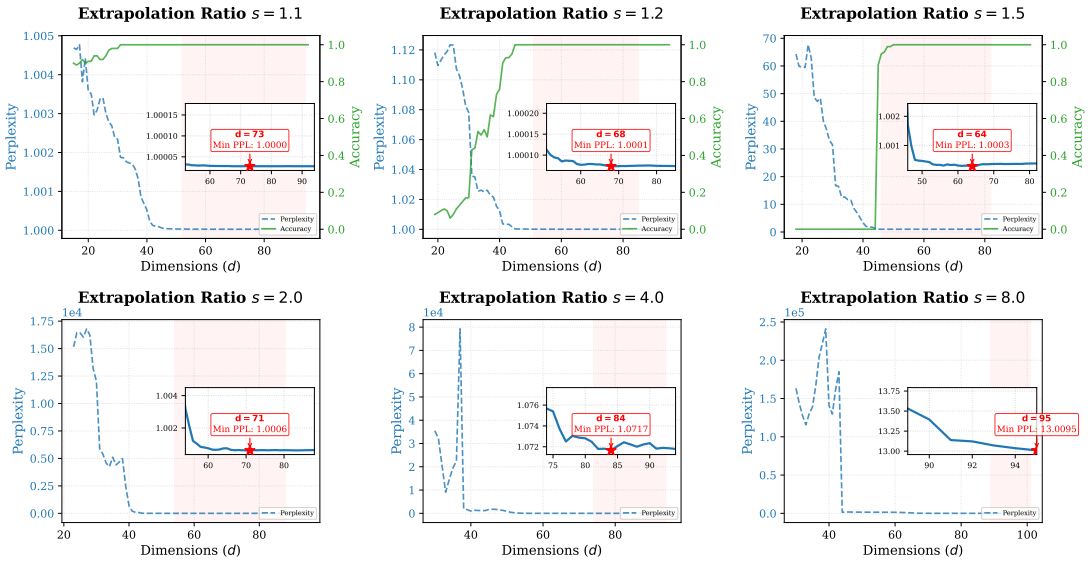


Figure 2: Inclusive-based results of PPL and Accuracy for various extrapolation ratios s for 100-digit string-copying task. Perplexity (PPL) as a function of the interpolation ending index d. The x-axis d represents the lower bound of the dimension range $[d_{\text{upper}}, d]$ where interpolation is applied (Equation 6). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

now partition the entire RoPE spectrum into three functional regions relative to their impact on length extrapolation:

- Harmful Range $[0, d_{\text{upper}})$: Dimensions that have higher frequencies than the Critical Band. This set of dimensions should not be interpolated. As these dimensions have higher frequencies, they are responsible for the local information. Therefore, interpolating these dimensions causes the model to confuse tokens that are relatively close in positional distance, leading to a decrease in model performance.
- Critical Band $[d_{\text{upper}}, d_{\text{lower}}]$: Dimensions that should be interpolated to maintain the extrapolation capability.
- Benign Range $(d_{\text{lower}}, d_{\text{max}}/2 - 1]$: Dimensions that have lower frequencies than the Critical Band. These dimensions are performance-neutral because their phase shifts during extrapolation are negligible; the model is largely indifferent to whether they are interpolated.

4.2 The Dynamics: Sliding Across Spectrum

After defining the boundaries of the Critical Band, we further explore its dynamic characteristics as the extrapolation ratio s varies. This "sliding" property is key to explaining the model's extrapolation behavior.

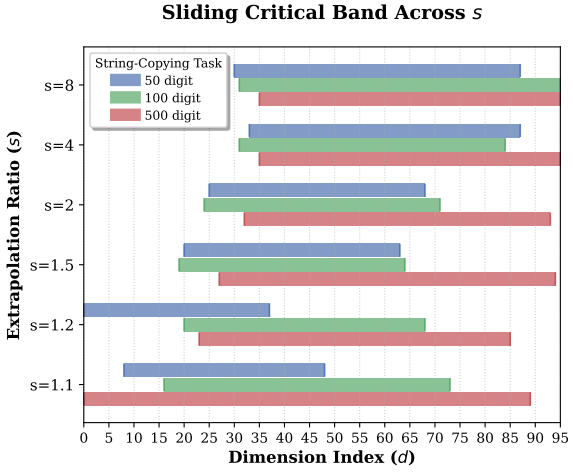


Figure 3: Sliding Critical Band across different task lengths and extrapolation ratios.

According to the different string-copying length settings (50, 100, 500-digit) as summarized in Table 2, the upper bound (d_{upper}) and lower bound (d_{lower}) of the Critical Band show a clear trend of shifting to the lower frequencies as the extrapolation ratio s increases. This sliding characteristic, visually demonstrated in Figure 3, reveals a hidden complexity in the relationship between frequency sensitivity and extrapolation. The functional roles of each rotary dimension are not fixed but exist in a dynamic state.

More importantly, this discovery refines the com-

Table 2: Critical band slides across different extrapolation ratios (s) for 50, 100, and 500-digit tasks.

Setting	$\mathbf{s = 1.1}$	$\mathbf{s = 1.2}$	$\mathbf{s = 1.5}$	$\mathbf{s = 2}$	$\mathbf{s = 4}$	$\mathbf{s = 8}$
50 digit	$[8, 48]$	$[0, 37]$	$[20, 63]$	$[25, 68]$	$[33, 87]$	$[30, 87]$
100 digit	$[16, 73]$	$[20, 68]$	$[19, 64]$	$[24, 71]$	$[31, 84]$	$[31, 95]$
500 digit	$[0, 89]$	$[23, 85]$	$[27, 94]$	$[32, 93]$	$[35, 95]$	$[35, 95]$

mon belief (Liu et al., 2024; Peng et al., 2023) that extrapolation failure is primarily caused by dimensions that have not completed a full rotation $d \geq d_{\text{extra}}$. Our analysis demonstrates that the maintenance of extrapolation capability does not depend on the completion of periodic rotations or the complexity of interpolation functions. Instead, it hinges entirely on whether the interpolation operation can precisely cover the Sliding Critical Band as it shifts. This insight identifies that the dynamic motion of the spectrum is a key factor affecting model performance, and it provides a viable direction for designing adaptive position encoding that can track these frequency boundaries in real time.

5 Spectrum Bandwidth Exhaustion

The dynamic migration of the Sliding Critical Band raises a fundamental question:

what happens when the extrapolation ratio s increases to a point where the required lower bound d_{lower} exceeds the maximum RoPE-dimension $d_{\text{max}}/2$ of the model?

In this section, we define this state as ***Spectrum Bandwidth Exhaustion***. This concept explains the ultimate collapse of extrapolation performance and also illuminates the underlying mechanism of why increasing the RoPE-base, a strategy adopted by Llama 3 (Grattafiori et al., 2024) and Code Llama (Rozière et al., 2024), is effective, and further supports Men et al. (2024)'s point of view.

As the extrapolation ratio s increases, the Critical Band slides to low-frequency dimensions. When the extrapolation ratio becomes extremely large, d_{lower} will eventually be larger than $d_{\text{max}}/2$, leading to spectrum overflow. This missing information is critical for maintaining extrapolation stability; the model will experience a sudden shift from performance degradation to model collapse. The purpose of increasing the RoPE-base is to lower the frequency floor. The frequency θ_d at the d-th dimension will then become smaller under larger RoPE-base θ compared to the original one. Therefore, the Sliding Critical Band, which would have exceeded $d_{\text{max}}/2$ when $\theta = 10,000$, is "pulled" back into the physical dimension space when $\theta = 100,000$. Moreover, the PPL also becomes smaller, as shown in Figure 4 and Table 3.

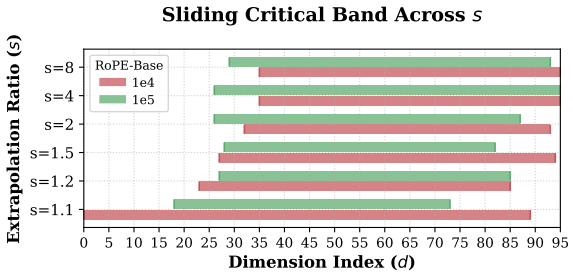


Figure 4: Sliding Critical Band for different RoPE bases with the same task length 500-digit.

Table 3: PPL results after interpolating critical band. Under large-scale extrapolation, larger RoPE-base achieves better performance.

s	$\theta = 10^4$	$\theta = 10^5$
$s = 2$	1.0013	1.0002
$s = 4$	1.0389	1.0178
$s = 8$	3.2016	1.9984

In summary, Spectrum Bandwidth Exhaustion represents the hard physical limit of RoPE-based models; the finiteness of the dimension spectrum ensures an eventual collapse. Enlarging the RoPE-base is, in essence, an act of 'buying more space' for the band to slide, thereby extending the functional horizon of the model's long-context extrapolation capabilities.

6 Recovering Slash Pattern Attention

In this section, we will discuss the necessity and effectiveness of interpolating Critical Band for enhancing the model's length extrapolation capability from the perspective of attention scores pattern. Cheng et al. (2026) proposed Slash-Dominant Heads (SDHs), a phenomenon where the attention

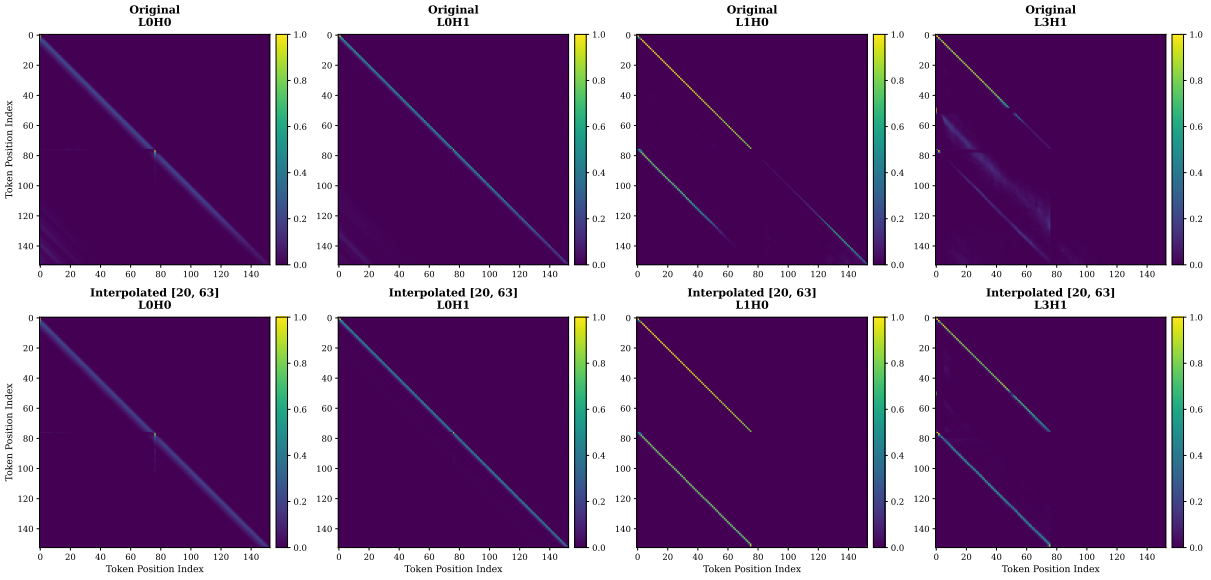


Figure 5: Comparison of attention pattern between original RoPE (first row) and Critical Band interpolation RoPE (second row) in 100-digit string-copying task with extrapolation ratio $s = 1.5$. The a-th head at b-th layer is denoted as LbHa. The slash attention pattern in the first row is weaker than that in the second row.

distribution is primarily determined by RoPE and is largely unaffected by semantic content. The slash patterns play an important role in long-context inference (Xu et al., 2025; Jiang et al., 2024).

As shown in the first row of Figure 5, when the model processes length-OOD sequences, its slash pattern tendency weakens, becomes chaotic, or even disappears. This indicates that the relative distances between tokens become blurred and disordered, which we believe is the underlying reason for the model's failure during length extrapolation. Critical Band interpolation restores the structural slash attention pattern (Figure 5), thereby reinstating the model's capacity to capture relative positional dependencies between tokens.

7 Validation in Real-World Scenarios

To further confirm the generalizability of the Sliding Critical Band, we conduct our experiments on LlaMA-7B (Touvron et al., 2023), specifically using the model weights adapted for the Hugging Face Transformers library (HuggyLLaMA, 2023). The dataset for evaluation is the English subset of C4 dataset (Raffel et al., 2023).

LlaMA-7B is pretrained with a $L_{\text{train}} = 2048$ context window length. We consider four extrapolation ratios: $s = 1.125$ ($L' = 2304$), $s = 1.25$ ($L' = 2560$), $s = 1.75$ ($L' = 3584$) and $s = 2.0$ ($L' = 4096$) . The C4 dataset is truncated to this desired test length. Following the procedure in Section 4.1, we find the Critical Bands for $s = 1.125, s = 1.25, s = 1.75, s = 2.0$, as illustrated in Figure 6. Remarkably, the Critical Band exhibits an identical rightward migration as the extrapolation ratio s increases, shifting from the higher-frequency region at $s = 1.125$ toward the lower-frequency dimensions at $s = 2$. This further validation on real-world model and natural language dataset suggests that sliding dynamics are a fundamental property of RoPE-based models. We provide the detailed results and convergence plots in Figure 13, Appendix B.

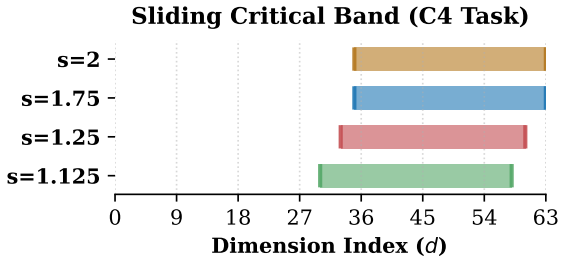


Figure 6: Sliding Critical Bands for LlaMA-7B on C4 with different extrapolation ratios.

To quantitatively evaluate the effectiveness of Sliding Critical Band mechanism on natural language modeling, we compare our approach against standard and static positional interpolation baselines on the C4 dataset. As shown in Table 4, we compared the perplexity of the Vanilla formulation, Positional Interpolation (Chen et al., 2023),

Table 4: Perplexity evaluation on the English C4 dataset across various context extrapolation ratios. We compare the Vanilla model, Position Interpolation (PI), YaRN, and our proposed Sliding Critical Band (SCB). For each method we report results both zero-shot (ZS, without fine-tuning) and after fine-tuning with targeted sequence length for 100 steps (FT). The best (lowest) perplexity in each column for a given extrapolation ratio is bolded. The reported perplexity is the mean of three different runs.

Ratio	Method	Evaluation Context Length			
		Original (2k)		Original × Ratio	
		ZS	FT	ZS	FT
1.125	Vanilla	6.0490	6.0533	6.7351	6.4862
	PI	6.0473	6.0381	6.4362	6.4271
	YaRN	6.0434	6.0385	6.4319	6.4245
	SCB (ours)	**6.0394**	**6.0374**	**6.4306**	**6.4241**
1.25	Vanilla	6.0490	6.0928	10.5324	7.2228
	PI	6.0596	6.0388	6.6323	6.6041
	YaRN	6.0559	6.0395	6.6147	6.6010
	SCB (ours)	**6.0434**	**6.0374**	**6.6110**	**6.5996**
1.75	Vanilla	6.0490	6.0906	59.4537	14.0116
	PI	6.2922	6.0502	5.9608	5.6688
	YaRN	6.0991	**6.0407**	5.7068	**5.6422**
	SCB (ours)	**6.0886**	6.0547	**5.6999**	5.6448
2.0	Vanilla	6.0490	6.1238	109.7982	19.1655
	PI	6.4359	6.6526	6.0521	6.2305
	YaRN	6.1227	**6.6136**	5.6790	**6.1631**
	SCB (ours)	**6.1212**	6.6611	**5.6637**	6.1643

YaRN (Peng et al., 2023), and our proposed Sliding Critical Band (SCB) across multiple extrapolation ratios. The evaluations were conducted under both zero-shot scenarios (without fine-tuning) and resource-efficient fine-tuning scenarios.

The empirical results demonstrate that our SCB consistently delivers highly competitive performance across different settings. Notably, SCB achieves the lowest perplexity across all extended-context scenarios in zero-shot configurations, demonstrating that the Sliding Critical Band is the intrinsic mechanism of the pre-trained model. We argue that zero-shot performance serves as a particularly rigorous validation of length extrapolation methods, as it rules out confounding effects from parameter updates. As illustrated in Table 4, even the Vanilla baseline exhibits a dramatic reduction in perplexity after only 100 steps of fine-tuning, from 109.7982 to 19.1655 at the 2.0 extension scale. This pervasive performance gain across all models during the adaptation phase underscores a critical insight: while advanced interpolation schemes like YaRN (Peng et al., 2023), with a small amount of fine-tuning, are traditionally used in practice, the Sliding Critical Band achieves competitive performance without incurring additional training costs.

8 Conclusion

In conclusion, our paper moves beyond the static view of RoPE extrapolation by revealing the inherent dynamic nature of spectral interpolation. We demonstrate that the requirements for interpolation are not fixed but instead evolve according to a Sliding Critical Band as the extrapolation scale increases. This framework provides a unified explanation of the effectiveness of existing methods and identifies the precise spectral dimensions responsible for stability across varying context lengths. Furthermore, we elucidate the mechanism by which increasing the RoPE base frequency enhances extrapolation, characterizing it as a strategy to delay Spectrum Bandwidth Exhaustion. The concepts of the Critical Band and Spectrum Bandwidth Exhaustion introduced in this work offer a novel perspective, which may provide insights for future research on context extension.

9 Limitations

While our work uncovers the dynamic nature of the Sliding Critical Band in RoPE-based models,

we acknowledge several limitations that provide avenues for future research.

First, our current empirical validation primarily focuses on perplexity metrics using the C4 dataset and exact-match accuracy on synthetic string-copying tasks. While perplexity is a reliable continuous proxy for language modeling capabilities, evaluating SCB on complex downstream long-context benchmarks involving multi-document QA and summarization would provide a more comprehensive assessment of its retrieval and reasoning capacities.

Second, due to limited computational resource, our experiments are conducted on custom 7M-parameter models and the LLaMA-7B architecture. Although the sliding dynamics are fundamental to the RoPE formulation itself, validating whether this exact frequency migration pattern perfectly scales to 70B+ parameter models or Mixture-of-Experts (MoE) architectures remains an important next step, as immense parameter scales may alter internal attention distributions.

Finally, future work may involve predicting the Sliding Critical Band based on model configurations and pre-training parameters (such as head dimension, RoPE base, pre-training context length, and extrapolation ratio).

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A Experimental Details

Training and Evaluation. All the models are trained from scratch using standard next-token prediction objective and cross-entropy loss, except the LlaMA-7B in Section 7 in a single A100 GPU. We compute the cross-entropy loss over the entire sequence; masking is applied to the padding tokens to ensure they do not contribute to the gradient updates. Training details are as follows:

- We use AdamW optimizer with betas $\beta = (0.9, 0.99)$ and epsilon $\epsilon = 1e - 12$. Weight decay is fixed to 0.1.
- We use linear warmup and cosine decay as the learning rate schedule.
- We do not use dropout.

B More Results

Table 5: Training Details and Hyperparameters

Task	Total Steps	Warmup	Decay	LR	# D_{train}	Batch size	Block size
50-digit	9000	1000	2000	$5e-4$	2M	1000	128
100-digit	9000	1000	2000	$5e-4$	3M	1000	256
500-digit	9000	1000	2000	$5e-4$	3M	800	1024
500-digit, θ=1e5	9000	1000	2000	$5e-4$	3M	800	1024

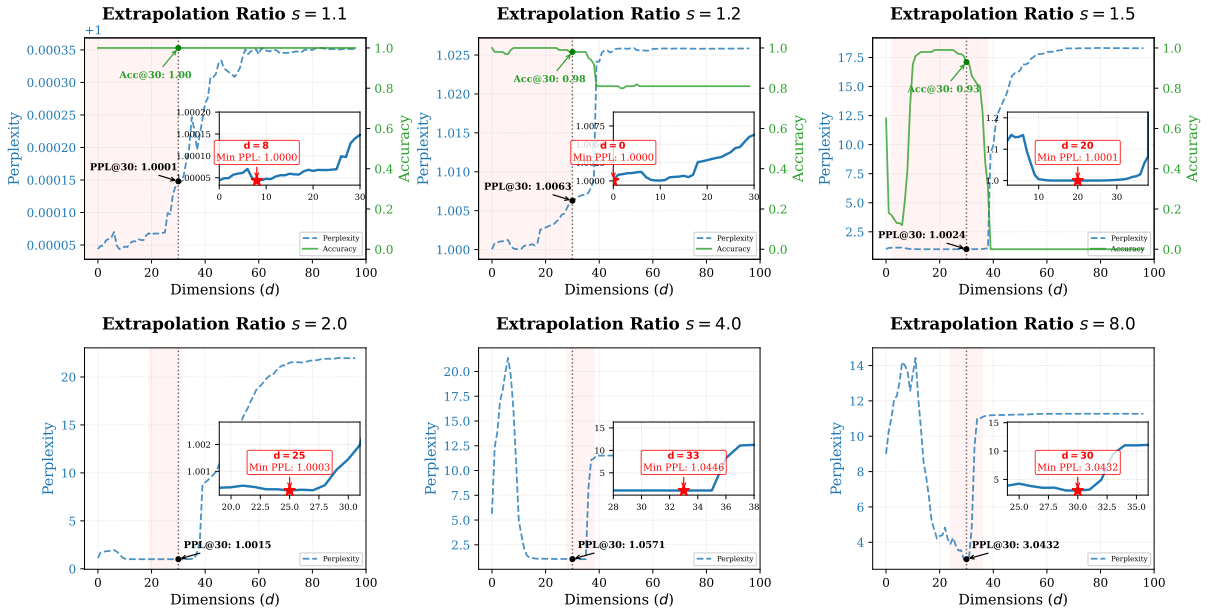


Figure 7: Exclusive-based results of PPL and Accuracy for various extrapolation ratios s for 50-digit string-copying task. Perplexity (PPL) as a function of the interpolation starting index d. The x-axis d represents the upper bound of the dimension range $[d, d_{\text{max}}/2 - 1]$ where interpolation is applied (Equation 5). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

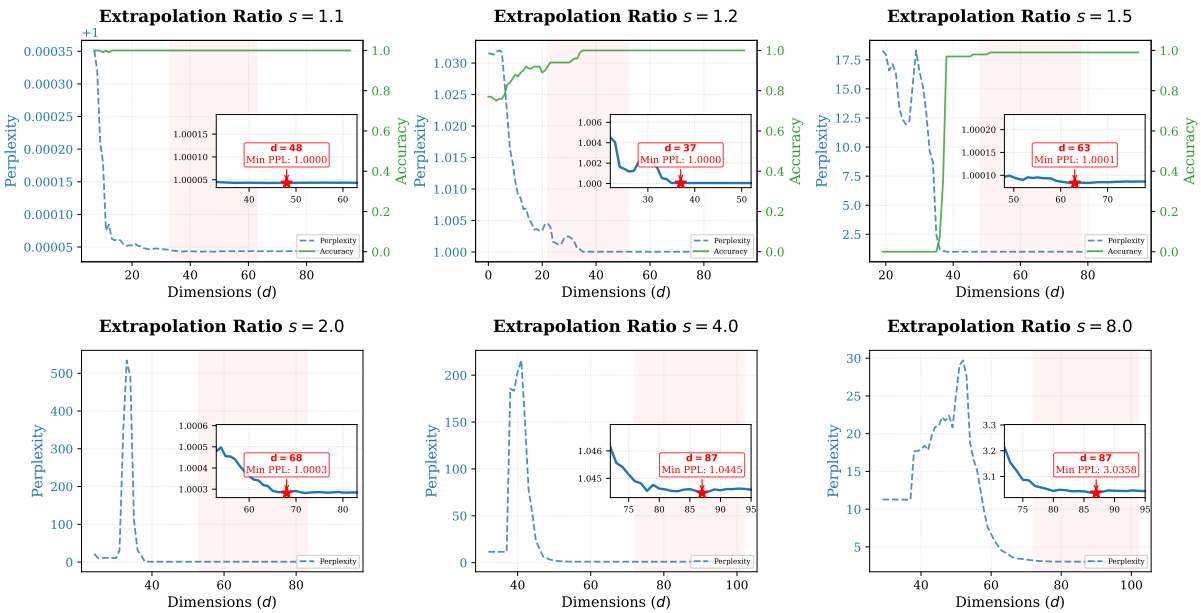


Figure 8: Inclusive-based results of PPL and Accuracy for various extrapolation ratios s for 50-digit string-copying task. Perplexity (PPL) as a function of the interpolation ending index d. The x-axis d represents the lower bound of the dimension range $[d_{\text{upper}}, d]$ where interpolation is applied (Equation 6). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

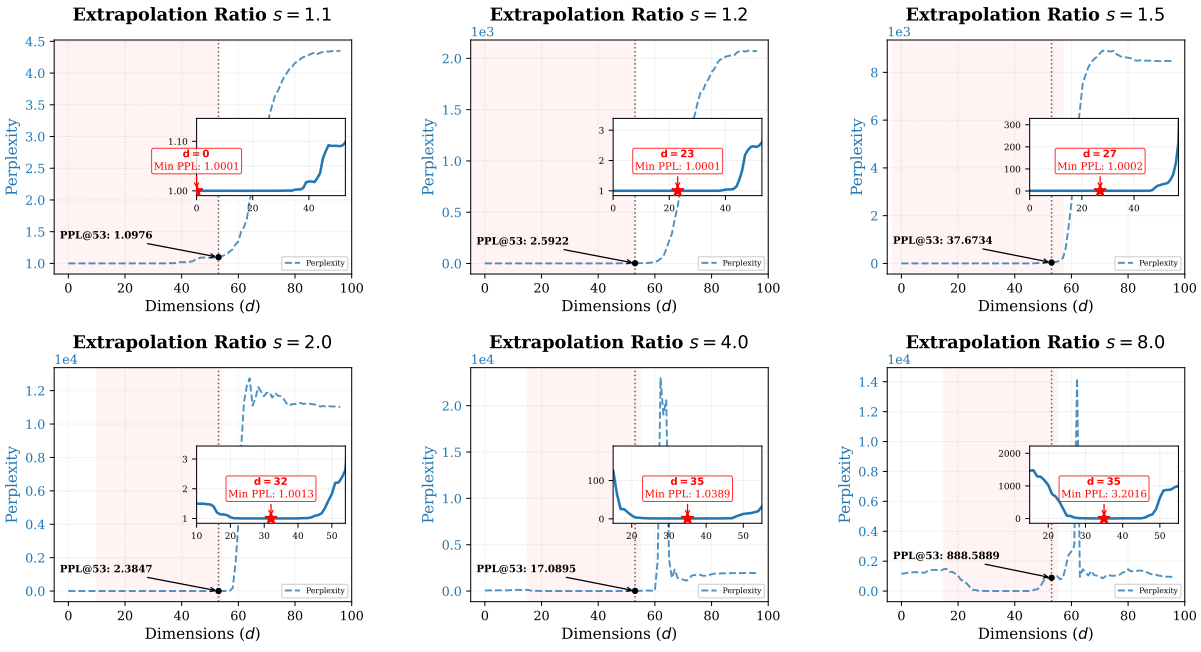


Figure 9: Exclusive-based results of PPL for various extrapolation ratios s for 500-digit string-copying task. Perplexity (PPL) as a function of the interpolation starting index d. The x-axis d represents the upper bound of the dimension range $[d, d_{\text{max}}/2 - 1]$ where interpolation is applied (Equation 5). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

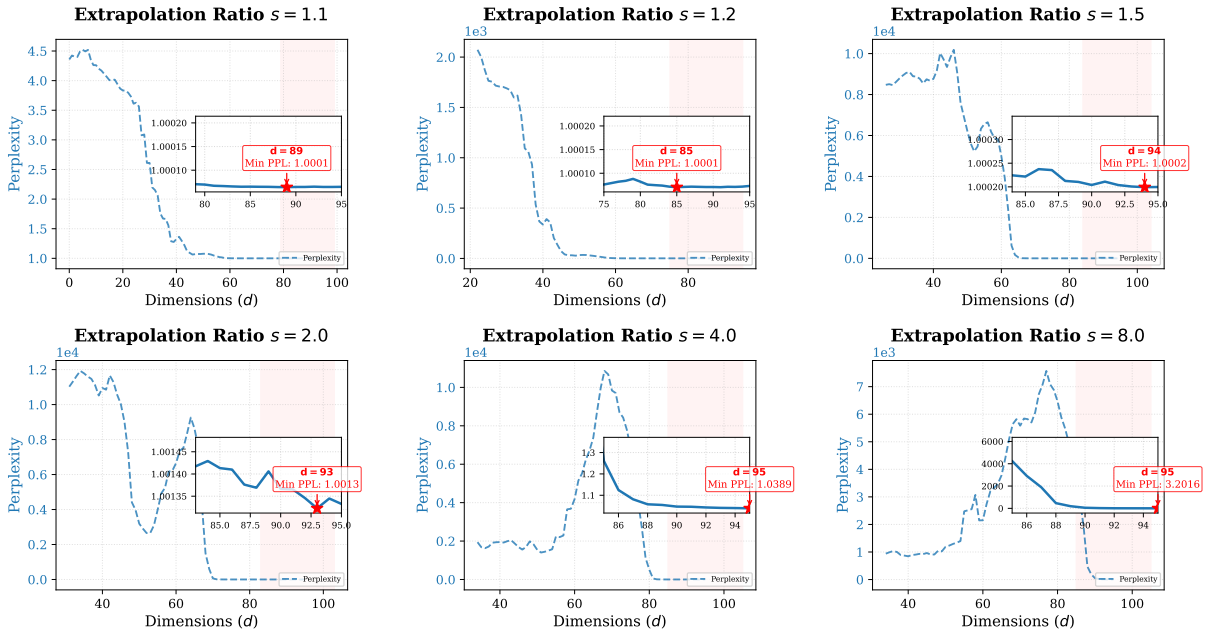


Figure 10: Inclusive-based results of PPL for various extrapolation ratios s for 500-digit string-copying task. Perplexity (PPL) as a function of the interpolation ending index d. The x-axis d represents the lower bound of the dimension range $[d_{\text{upper}}, d]$ where interpolation is applied (Equation 6). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

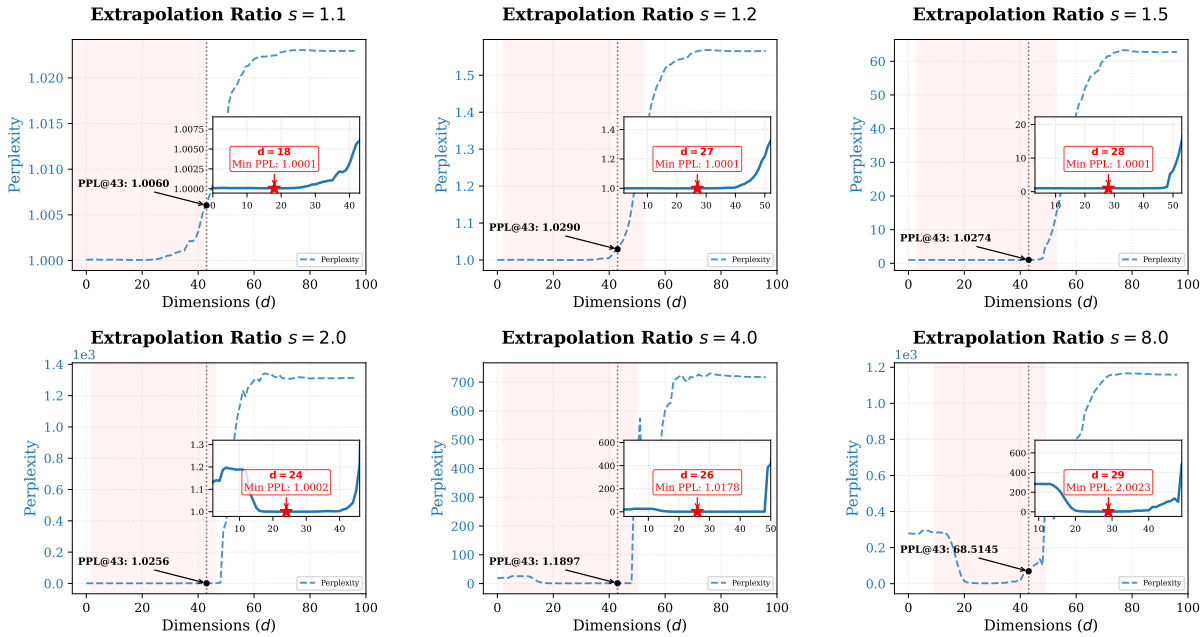


Figure 11: Exclusive-based results of PPL for various extrapolation ratios s for 500-digit string-copying task with RoPE-base $\theta = 100,000$. Perplexity (PPL) as a function of the interpolation starting index d. The x-axis d represents the upper bound of the dimension range $[d, d_{\text{max}}/2 - 1]$ where interpolation is applied (Equation 5). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

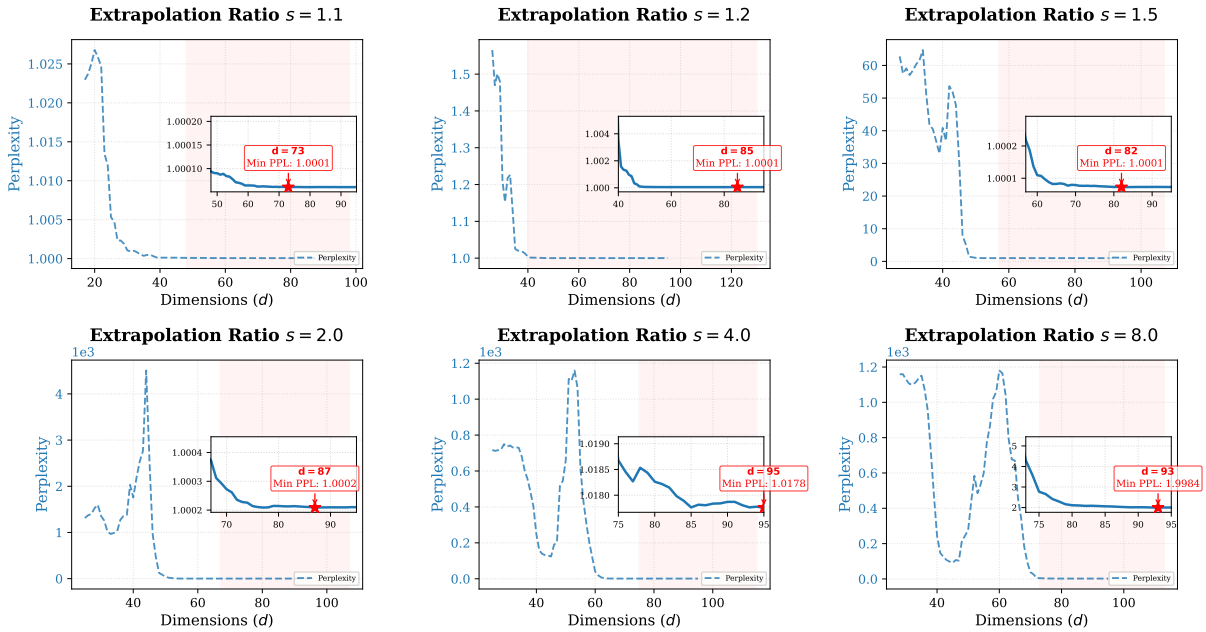


Figure 12: Inclusive-based results of PPL for various extrapolation ratios s for 500-digit string-copying task with RoPE-base $\theta = 100,000$. Perplexity (PPL) as a function of the interpolation ending index d. The x-axis d represents the lower bound of the dimension range $[d_{\text{upper}}, d]$ where interpolation is applied (Equation 6). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

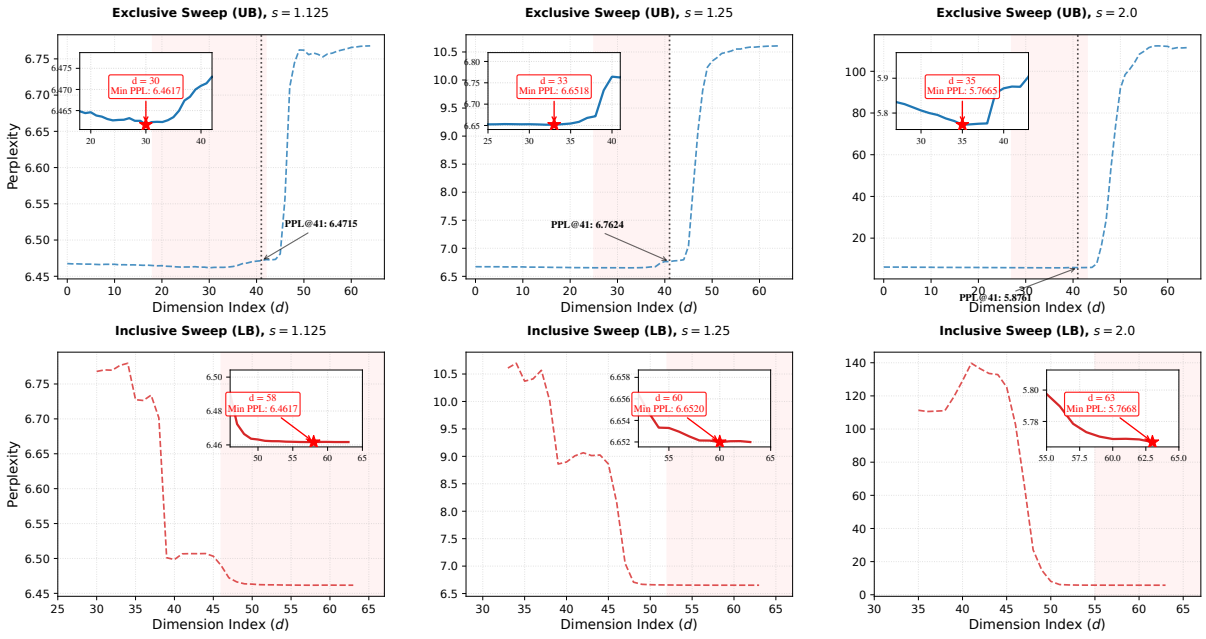


Figure 13: PPL for various extrapolation ratios s for LlaMA-7B on C4 dataset. Perplexity (PPL) as a function of the interpolation index d. For the first row, x-axis d represents the upper bound of the dimension range $[d, d_{\text{max}}/2 - 1]$ where interpolation is applied (Equation 5). For the second row, x-axis d represents the lower bound of the dimension range $[d_{\text{upper}}, d]$ where interpolation is applied (Equation 6). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.